Exploratory Data Analysis Report

Dataset: M1_combined_monthly_raw.csv
Generated: 2026-10-18T09:12:28.931015+00:00
Date-like columns: Date

Rows	Columns	Numeric	Categorical/date
302	77	65	12

Duplicate rows
2

Analysis scope

Target status: TB_Total_Exports This report describes observed data only; it does not clean, transform, or overwrite the source dataset.

Confirmed target analysis

TB_Total_Exports is the confirmed target for this report. Correlations are observational associations only; they do not establish causation or forecast performance.

Target descriptive statistics

column	count	mean	std	min	1%	5%	25%	50%	75%	95%	99%
TB_Total_Exports	296	2.024e+07	2.568e+07	18.35	20.21	22.97	1.518e+07	2.277e+07	2.528e+07	2.812e+07	4.093e+07

Numerical feature relationships with the target

feature	correlation	absolute_correlation
CPI_base_2006_07	0.5567	0.5567
TB_Tea_Bags	0.477	0.477

feature	correlation	absolute_correlation
TB_Instant_Tea	0.4734	0.4734
TB_Packets	0.4673	0.4673
TB_Green_Tea	0.4631	0.4631
TB_Bulk_Tea	0.4595	0.4595
CT_World_value_USD	0.4333	0.4333
GDP_Syrian_Arab_Republic	-0.4271	0.4271
CT_kg_Iraq	0.3173	0.3173
TB_Export_Value_Rs_Mn	0.3047	0.3047
COMP_Kenya	0.297	0.297
CT_kg_Syrian_Arab_Republic	-0.2535	0.2535
INF_Iraq	0.2324	0.2324
AUC_Low_price_mean	0.2287	0.2287
AUC_High_price_mean	0.2272	0.2272
COMP_India	0.2111	0.2111
AUC_Medium_price_mean	0.2041	0.2041
CT_kg_Turkiye	0.2032	0.2032
INF_Libya	-0.2009	0.2009
FX_USD_LKR	0.1971	0.1971
INF_United_Arab_Emirates	0.1865	0.1865
INF_Turkiye	-0.176	0.176
INF_Syrian_Arab_Republic	0.1752	0.1752
CT_kg_Libya	-0.1723	0.1723
COMP_China	0.1555	0.1555

Data quality overview

column	dtype	role	rows	missing_count	missing_pct	unique_values_including_missing	c
Date	object	date-like	302	0	0	300	0
TB_Bulk_Tea	float64	numeric	302	3	0.993	297	0
TB_Packets	float64	numeric	302	3	0.993	298	0

column	dtype	role	rows	missing_count	missing_pct	unique_values_including_missing	c
TB_Tea_Bags	float64	numeric	302	3	0.993	294	0
TB_Instant_Tea	float64	numeric	302	3	0.993	288	0
TB_Green_Tea	float64	numeric	302	3	0.993	274	0
TB_Total_Exports	float64	numeric	302	6	1.987	294	0
TB_Quantity_Unit	object	categorical	302	0	0	2	0
TB_Export_Value_Rs_Mn	float64	numeric	302	8	2.649	293	0
CT_World_netWgt	float64	numeric	302	7	2.318	294	0
CT_World_unit	object	categorical	302	7	2.318	3	0
CT_World_value_USD	float64	numeric	302	17	5.629	284	0
CT_kg_Azerbaijan	float64	numeric	302	15	4.967	286	0
CT_kg_Chile	float64	numeric	302	19	6.291	282	0
CT_kg_Iran	float64	numeric	302	14	4.636	287	0
CT_kg_Iraq	float64	numeric	302	11	3.642	290	0
CT_kg_Japan	float64	numeric	302	11	3.642	290	0
CT_kg_Libya	float64	numeric	302	14	4.636	287	0
CT_kg_Russian_Federation	float64	numeric	302	18	5.96	283	0
CT_kg_Syrian_Arab_Republic	float64	numeric	302	11	3.642	290	0

column	dtype	role	rows	missing_count	missing_pct	unique_values_including_missing	c
CT_kg_Turkiye	float64	numeric	302	11	3.642	290	0
CT_kg_United_Arab_Emirates	float64	numeric	302	9	2.98	292	0
AUC_n_sale_weeks_held	int64	numeric	302	0	0	5	0
AUC_n_sale_weeks_no_sale	int64	numeric	302	0	0	4	0
AUC_n_rows_price_in_USD	int64	numeric	302	0	0	3	0
AUC_n_rows_qty_in_MT	int64	numeric	302	0	0	3	0
AUC_n_held_rows_price_missing	int64	numeric	302	0	0	5	0
AUC_High_price_mean	float64	numeric	302	1	0.331	300	0
AUC_High_price_std	float64	numeric	302	1	0.331	300	0
AUC_High_qty_sum	float64	numeric	302	1	0.331	300	0
AUC_Medium_price_mean	float64	numeric	302	2	0.662	299	0
AUC_Medium_price_std	float64	numeric	302	2	0.662	299	0
AUC_Medium_qty_sum	float64	numeric	302	1	0.331	300	0
AUC_Low_price_mean	float64	numeric	302	1	0.331	300	0
AUC_Low_price_std	float64	numeric	302	1	0.331	300	0
AUC_Low_qty_sum	float64	numeric	302	1	0.331	300	0
ONI_season	object	categorical	302	0	0	12	0
ONI_total	float64	numeric	302	0	0	200	0

column	dtype	role	rows	missing_count	missing_pct	unique_values_including_missing	c
ONI_anom	float64	numeric	302	0	0	176	0
RAIN_Badulla	float64	numeric	302	13	4.305	269	0
RAIN_Nuwara_Eliya	float64	numeric	302	13	4.305	280	0
RAIN_Ratnapura	float64	numeric	302	8	2.649	285	0
RAIN_unit_Badulla	object	categorical	302	0	0	2	0
RAIN_unit_Nuwara_Eliya	object	categorical	302	0	0	2	0
RAIN_unit_Ratnapura	object	categorical	302	0	0	2	0
FX_USD_LKR	float64	numeric	302	0	0	300	0
FX_origin	object	categorical	302	0	0	2	0
CPI_base_2006_07	float64	numeric	302	123	40.73	178	0
CPI_base_2013	float64	numeric	302	158	52.32	142	0
COMP_China	float64	numeric	302	12	3.974	287	0
COMP_India	float64	numeric	302	14	4.636	284	0
COMP_Kenya	float64	numeric	302	12	3.974	286	0
COMP_Vietnam	float64	numeric	302	12	3.974	286	0
COMP_unit_China	object	categorical	302	0	0	1	1

column	dtype	role	rows	missing_count	missing_pct	unique_values_including_missing	c
COMP_unit_India	object	categorical	302	0	0	1	1
COMP_unit_Kenya	object	categorical	302	0	0	1	1
COMP_unit_Vietnam	object	categorical	302	0	0	2	0
GDP_Azerbaijan	float64	numeric	302	12	3.974	25	0
GDP_Chile	float64	numeric	302	0	0	25	0
GDP_Iran	float64	numeric	302	12	3.974	25	0
GDP_Iraq	float64	numeric	302	48	15.89	22	0
GDP_Japan	float64	numeric	302	0	0	23	0
GDP_Libya	float64	numeric	302	24	7.947	24	0
GDP_Russian_Federation	float64	numeric	302	0	0	24	0
GDP_Syrian_Arab_Republic	float64	numeric	302	156	51.66	13	0
GDP_Turkiye	float64	numeric	302	12	3.974	25	0
GDP_United_Arab_Emirates	float64	numeric	302	12	3.974	25	0

column	dtype	role	rows	missing_count	missing_pct	unique_values_including_missing	c
INF_Azerbaijan	float64	numeric	302	12	3.974	25	0
INF_Chile	float64	numeric	302	0	0	24	0
INF_Iran	float64	numeric	302	24	7.947	24	0
INF_Iraq	float64	numeric	302	0	0	24	0
INF_Japan	float64	numeric	302	0	0	23	0
INF_Libya	float64	numeric	302	12	3.974	23	0
INF_Russian_Federation	float64	numeric	302	0	0	25	0
INF_Syrian_Arab_Republic	float64	numeric	302	156	51.66	13	0
INF_Turkiye	float64	numeric	302	12	3.974	24	0
INF_United_Arab_Emirates	float64	numeric	302	12	3.974	22	0

Missing values

column	missing_count	missing_pct	non_missing_count
CPI_base_2013	158	52.32	144
GDP_Syrian_Arab_Republic	156	51.66	146
INF_Syrian_Arab_Republic	156	51.66	146
CPI_base_2006_07	123	40.73	179
GDP_Iraq	48	15.89	254

column	missing_count	missing_pct	non_missing_count
GDP_Libya	24	7.947	278
INF_Iran	24	7.947	278
CT_kg_Chile	19	6.291	283
CT_kg_Russian_Federation	18	5.96	284
CT_World_value_USD	17	5.629	285
CT_kg_Azerbaijan	15	4.967	287
COMP_India	14	4.636	288
CT_kg_Iran	14	4.636	288
CT_kg_Libya	14	4.636	288
RAIN_Badulla	13	4.305	289
RAIN_Nuwara_Eliya	13	4.305	289
COMP_China	12	3.974	290
COMP_Kenya	12	3.974	290
COMP_Vietnam	12	3.974	290
GDP_Azerbaijan	12	3.974	290
GDP_Iran	12	3.974	290
GDP_Turkiye	12	3.974	290
GDP_United_Arab_Emirates	12	3.974	290
INF_Azerbaijan	12	3.974	290
INF_Libya	12	3.974	290
INF_Turkiye	12	3.974	290
INF_United_Arab_Emirates	12	3.974	290
CT_kg_Iraq	11	3.642	291
CT_kg_Japan	11	3.642	291
CT_kg_Syrian_Arab_Republic	11	3.642	291
CT_kg_Turkiye	11	3.642	291
CT_kg_United_Arab_Emirates	9	2.98	293
RAIN_Ratnapura	8	2.649	294
TB_Export_Value_Rs_Mn	8	2.649	294
CT_World_netWgt	7	2.318	295
CT_World_unit	7	2.318	295
TB_Total_Exports	6	1.987	296
TB_Bulk_Tea	3	0.993	299

column	missing_count	missing_pct	non_missing_count
TB_Green_Tea	3	0.993	299
TB_Instant_Tea	3	0.993	299
TB_Packets	3	0.993	299
TB_Tea_Bags	3	0.993	299
AUC_Medium_price_mean	2	0.662	300
AUC_Medium_price_std	2	0.662	300
AUC_High_price_mean	1	0.331	301
AUC_High_price_std	1	0.331	301
AUC_High_qty_sum	1	0.331	301
AUC_Low_price_mean	1	0.331	301
AUC_Low_price_std	1	0.331	301
AUC_Low_qty_sum	1	0.331	301
AUC_Medium_qty_sum	1	0.331	301
AUC_n_held_rows_price_missing	0	0	302
AUC_n_rows_price_in_USD	0	0	302
AUC_n_rows_qty_in_MT	0	0	302
AUC_n_sale_weeks_held	0	0	302
AUC_n_sale_weeks_no_sale	0	0	302
COMP_unit_China	0	0	302
COMP_unit_India	0	0	302
COMP_unit_Kenya	0	0	302
COMP_unit_Vietnam	0	0	302
Date	0	0	302
FX_USD_LKR	0	0	302
FX_origin	0	0	302
GDP_Chile	0	0	302
GDP_Japan	0	0	302
GDP_Russian_Federation	0	0	302
INF_Chile	0	0	302
INF_Iraq	0	0	302
INF_Japan	0	0	302
INF_Russian_Federation	0	0	302
ONI_anom	0	0	302

column	missing_count	missing_pct	non_missing_count
ONI_season	0	0	302
ONI_total	0	0	302
RAIN_unit_Badulla	0	0	302
RAIN_unit_Nuwara_Eliya	0	0	302
RAIN_unit_Ratnapura	0	0	302
TB_Quantity_Unit	0	0	302

Descriptive statistics

column	count	mean	std	min	1%	5%	25%	50%	75%
TB_Bulk_Tea	299	8.905e+06	5.219e+06	9.597	10.58	12.16	7.885e+06	1.098e+07	1.27e
TB_Packets	299	6.17e+06	3.568e+06	6.144	6.736	7.576	5.797e+06	7.734e+06	8.587
TB_Tea_Bags	299	2.349e+06	1.35e+06	1.976	2.085	2.41	2.263e+06	2.995e+06	3.248
TB_Instant_Tea	299	4.691e+05	2.697e+05	0.388	0.4289	0.493	4.298e+05	6.002e+05	6.509
TB_Green_Tea	299	1.801e+05	1.047e+05	0.189	0.1999	0.229	1.677e+05	2.267e+05	2.531
TB_Total_Exports	296	2.024e+07	2.568e+07	18.35	20.21	22.97	1.518e+07	2.277e+07	2.528
TB_Export_Value_Rs_Mn	294	1.477e+04	8,430	4,120	4,470	5,118	7,732	1.394e+04	1.819
CT_World_netWgt	295	2.298e+07	4.671e+06	1.815e+04	2.243e+04	1.761e+07	2.131e+07	2.385e+07	2.542
CT_World_value_USD	285	9.493e+07	2.44e+07	4.315e+07	4.995e+07	5.615e+07	7.319e+07	9.966e+07	1.125
CT_kg_Azerbaijan	287	1.025e+06	1.715e+05	5.141e+05	6.34e+05	7.445e+05	9.148e+05	1.026e+06	1.135
CT_kg_Chile	283	8.108e+05	1.385e+05	4.925e+05	5.344e+05	5.892e+05	7.137e+05	8.109e+05	9.103
CT_kg_Iran	288	1.276e+06	4.243e+05	3.744e+05	4.625e+05	5.727e+05	8.89e+05	1.382e+06	1.565
CT_kg_Iraq	291	2.93e+06	5.719e+05	1.617e+06	1.774e+06	2.042e+06	2.514e+06	2.884e+06	3.36e
CT_kg_Japan	291	1.252e+06	2.583e+05	6.117e+05	6.909e+05	8.364e+05	1.073e+06	1.26e+06	1.428
CT_kg_Libya	288	9.346e+05	2.208e+05	4.297e+05	4.543e+05	5.381e+05	7.886e+05	9.516e+05	1.076
CT_kg_Russian_Federation	284	2.416e+06	4.884e+05	9.471e+05	1.202e+06	1.485e+06	2.149e+06	2.439e+06	2.723
CT_kg_Syrian_Arab_Republic	291	6.809e+05	4.067e+05	1.93e+05	2.063e+05	2.39e+05	2.89e+05	4.355e+05	1.056
CT_kg_Turkiye	291	2.397e+06	3.937e+05	1.389e+06	1.582e+06	1.788e+06	2.129e+06	2.383e+06	2.647
CT_kg_United_Arab_Emirates	293	2.505e+06	3.905e+05	1.59e+06	1.694e+06	1.86e+06	2.242e+06	2.495e+06	2.753
AUC_n_sale_weeks_held	302	4.159	0.6218	0	3	3	4	4	5
AUC_n_sale_weeks_no_sale	302	0.1887	0.4547	0	0	0	0	0	0
AUC_n_rows_price_in_USD	302	0.1424	0.3861	0	0	0	0	0	0
AUC_n_rows_qty_in_MT	302	0.3013	0.5516	0	0	0	0	0	1

column	count	mean	std	min	1%	5%	25%	50%	75%
AUC_n_held_rows_price_missing	302	0.2053	0.4866	0	0	0	0	0	0
AUC_High_price_mean	301	646.1	450.5	154.7	181.6	207.7	316.7	550.4	798.7
AUC_High_price_std	301	41.99	174.3	1.275	2.547	3.831	8.079	15.16	25.62
AUC_High_qty_sum	301	4.705e+06	9.241e+05	2.562e+06	2.843e+06	3.21e+06	4.099e+06	4.731e+06	5.2e-
AUC_Medium_price_mean	300	500.9	381.9	116.9	129.7	155.1	241.2	422	601.8
AUC_Medium_price_std	300	56.44	299.5	1.142	1.491	2.676	6.316	10.96	21.12
AUC_Medium_qty_sum	301	2.566e+06	5.201e+05	8.243e+05	1.39e+06	1.763e+06	2.223e+06	2.569e+06	2.867
AUC_Low_price_mean	301	682	471.2	197.8	207.2	222.8	335.8	581.9	846.3
AUC_Low_price_std	301	30.3	70.18	0.753	1.449	3.104	8.082	14.85	26.73
AUC_Low_qty_sum	301	1.396e+07	2.869e+06	6.613e+06	7.166e+06	8.765e+06	1.197e+07	1.4e+07	1.568
ONI_total	302	27.01	0.8574	25.1	25.18	25.56	26.47	27.04	27.54
ONI_anom	302	0.006225	0.7854	-1.53	-1.46	-1.17	-0.595	0	0.467
RAIN_Badulla	289	126.4	187.1	-999	-999	29.28	62.2	109.7	201.4
RAIN_Nuwara_Eliya	289	207.6	266.8	-999	16.31	59.5	120.6	181	246.1
RAIN_Ratnapura	294	293.4	220.5	-999	-60.89	67.73	161.4	280.1	398.5
FX_USD_LKR	302	148.6	70.33	76.68	78.7	87.58	103.6	125.5	163.4
CPI_base_2006_07	179	116.3	47.33	51.9	52.16	54.27	71.8	107.8	158.2
CPI_base_2013	144	144.2	50.92	97.1	97.76	101	108.3	124.5	146.5
COMP_China	290	2.307e+04	4,152	1.453e+04	1.557e+04	1.704e+04	1.983e+04	2.247e+04	2.596
COMP_India	288	1.879e+04	2,432	1.328e+04	1.404e+04	1.505e+04	1.708e+04	1.862e+04	2.056
COMP_Kenya	290	3.164e+04	8,363	1.549e+04	1.683e+04	1.887e+04	2.479e+04	3.142e+04	3.857
COMP_Vietnam	290	1.231e+06	2.806e+06	3,114	3,469	3,965	5,710	9,264	1.184
GDP_Azerbaijan	290	4.467	3.274	-1.33	-1.33	-0.76	2.45	3.93	6.45
GDP_Chile	302	3.193	1.601	-1.24	-1.24	0.48	2.75	3.3	3.93
GDP_Iran	290	2.395	2.416	-3.31	-3.31	-1.33	0.43	2.5	4.49
GDP_Iraq	254	6.089	2.543	1.23	1.23	1.38	4.85	7.2	7.97
GDP_Japan	302	0.5708	1.332	-3.42	-3.42	-2.25	0.28	0.76	1.37
GDP_Libya	278	3.022	6.326	-9.94	-9.94	-4.14	-2.42	1.88	9.11
GDP_Russian_Federation	302	2.445	1.969	-1.82	-1.82	-1.2	0.94	2.72	3.73
GDP_Syrian_Arab_Republic	146	2.601	2.18	-0.26	-0.26	-0.26	1.27	1.61	3.33
GDP_Turkiye	290	4.353	1.885	1.36	1.36	1.76	2.69	4.55	6.035
GDP_United_Arab_Emirates	290	2.011	1.944	-1.39	-1.39	-0.72	0.28	1.86	3.35
INF_Azerbaijan	290	9.084	4.555	1.15	1.15	1.57	5.86	9.04	13.36

column	count	mean	std	min	1%	5%	25%	50%	75%
INF_Chile	302	4.028	2.463	0.1	0.1	0.48	2.93	3.78	5.09
INF_Iran	278	19.9	10.64	7.03	7.03	9.76	12.17	14.64	26.7
INF_Iraq	302	6.397	4.715	-1	-1	-1	3.53	6.28	7.57
INF_Japan	302	0.402	1.081	-1	-1	-0.87	-0.31	0.08	0.49
INF_Libya	290	7.613	4.681	-1	-1	-1	4.71	8.04	10.78
INF_Russian_Federation	302	8.899	4.962	0.33	0.33	0.42	5.9	8.4	11.23
INF_Syrian_Arab_Republic	146	5.457	2.899	0.86	0.86	0.86	3.74	5.98	7.232
INF_Turkiye	290	20.33	19.22	6.3	6.3	6.5	8.3	10.4	19.6
INF_United_Arab_Emirates	290	1.857	2.195	-1	-1	-1	0.0025	1.59	3.2

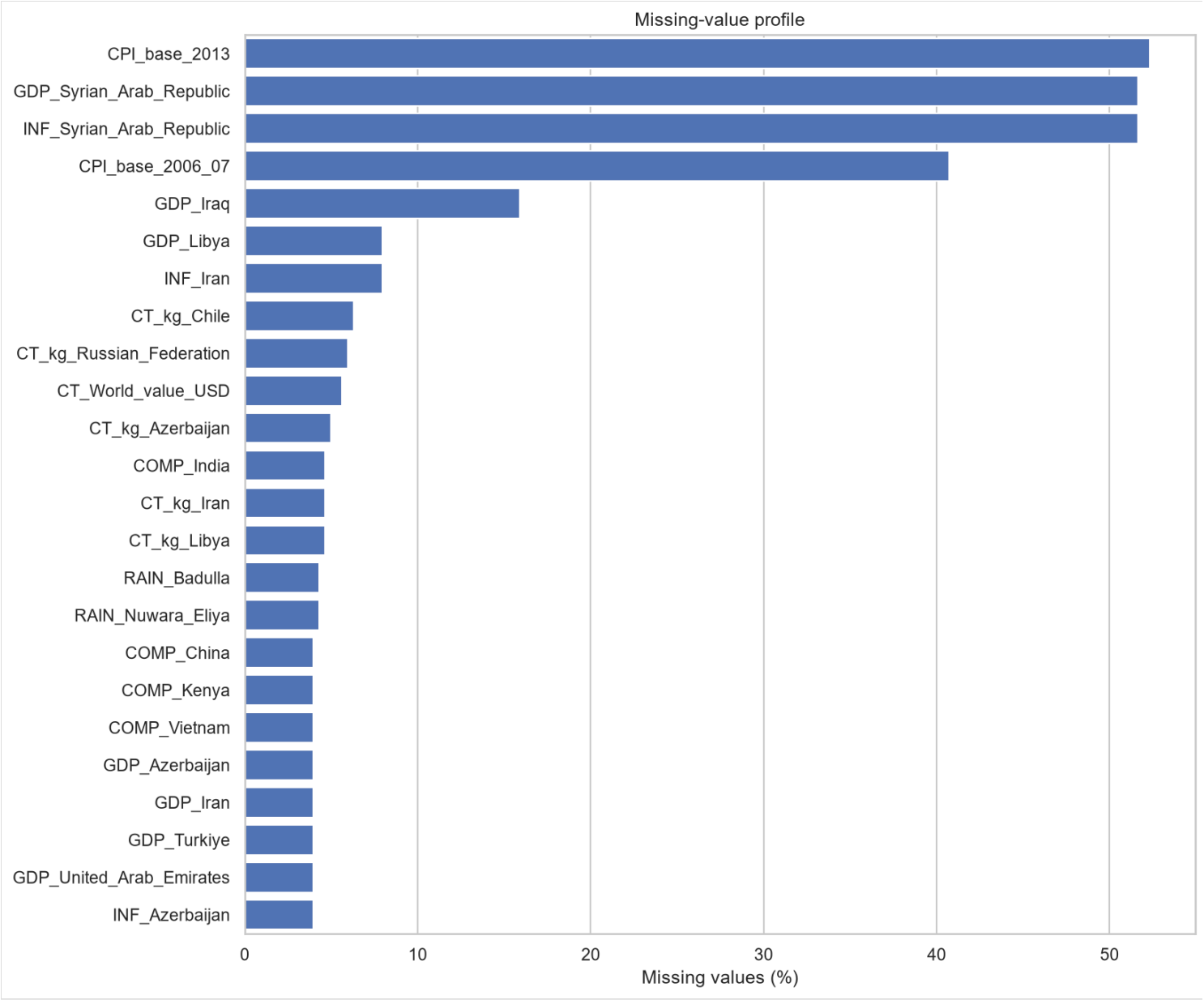
Strongest numerical relationships

feature_1	feature_2	correlation	absolute_correlation
CPI_base_2006_07	CPI_base_2013	1	1
TB_Tea_Bags	TB_Instant_Tea	0.9966	0.9966
TB_Packets	TB_Green_Tea	0.9963	0.9963
TB_Bulk_Tea	TB_Green_Tea	0.9947	0.9947
TB_Bulk_Tea	TB_Packets	0.9928	0.9928
TB_Packets	TB_Tea_Bags	0.9879	0.9879
TB_Packets	TB_Instant_Tea	0.9874	0.9874
TB_Tea_Bags	TB_Green_Tea	0.9844	0.9844
TB_Instant_Tea	TB_Green_Tea	0.9836	0.9836
AUC_High_price_mean	AUC_Low_price_mean	0.9803	0.9803
AUC_Low_price_mean	FX_USD_LKR	0.9794	0.9794
AUC_Low_price_mean	CPI_base_2006_07	0.9785	0.9785
TB_Bulk_Tea	TB_Instant_Tea	0.9727	0.9727
TB_Bulk_Tea	TB_Tea_Bags	0.9725	0.9725
AUC_Medium_price_mean	CPI_base_2006_07	0.9662	0.9662
AUC_High_price_mean	FX_USD_LKR	0.9649	0.9649
TB_Export_Value_Rs_Mn	AUC_Low_price_mean	0.9645	0.9645
TB_Export_Value_Rs_Mn	CPI_base_2006_07	0.9574	0.9574
FX_USD_LKR	CPI_base_2006_07	0.9548	0.9548

feature_1	feature_2	correlation	absolute_correlation
TB_Export_Value_Rs_Mn	AUC_High_price_mean	0.9494	0.9494
TB_Export_Value_Rs_Mn	FX_USD_LKR	0.9401	0.9401
CT_World_value_USD	CPI_base_2006_07	0.9367	0.9367
AUC_Low_price_mean	CPI_base_2013	0.928	0.928
ONI_total	ONI_anom	0.9105	0.9105
AUC_High_price_mean	CPI_base_2013	0.9096	0.9096
AUC_Medium_price_mean	AUC_Low_price_mean	0.9045	0.9045
TB_Export_Value_Rs_Mn	CPI_base_2013	0.9019	0.9019
FX_USD_LKR	CPI_base_2013	0.8969	0.8969
AUC_High_price_mean	AUC_Medium_price_mean	0.8908	0.8908
AUC_High_price_mean	CPI_base_2006_07	0.889	0.889

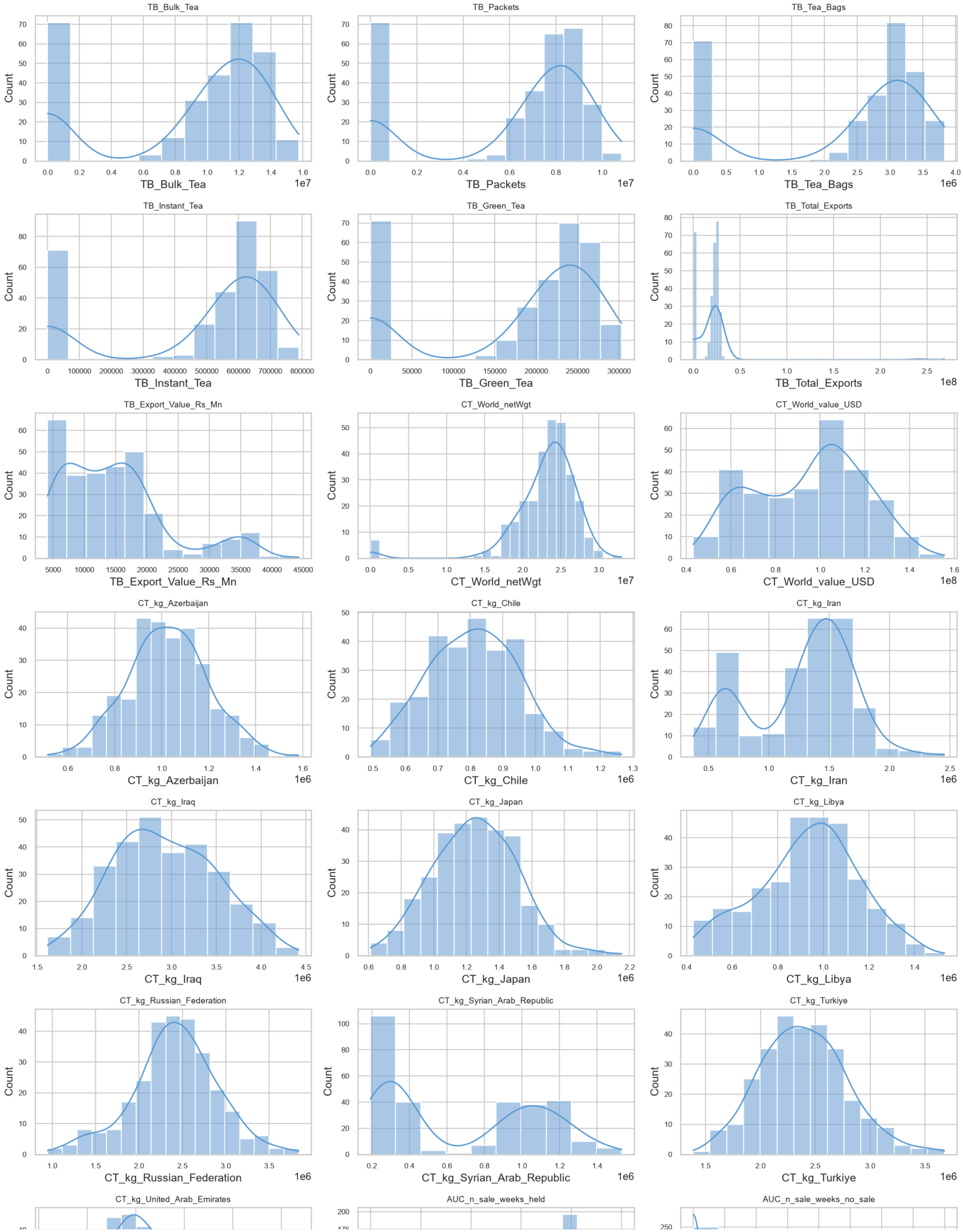
27 feature pairs have absolute correlation at least 0.90 and should be reviewed for multicollinearity or derived-variable duplication.

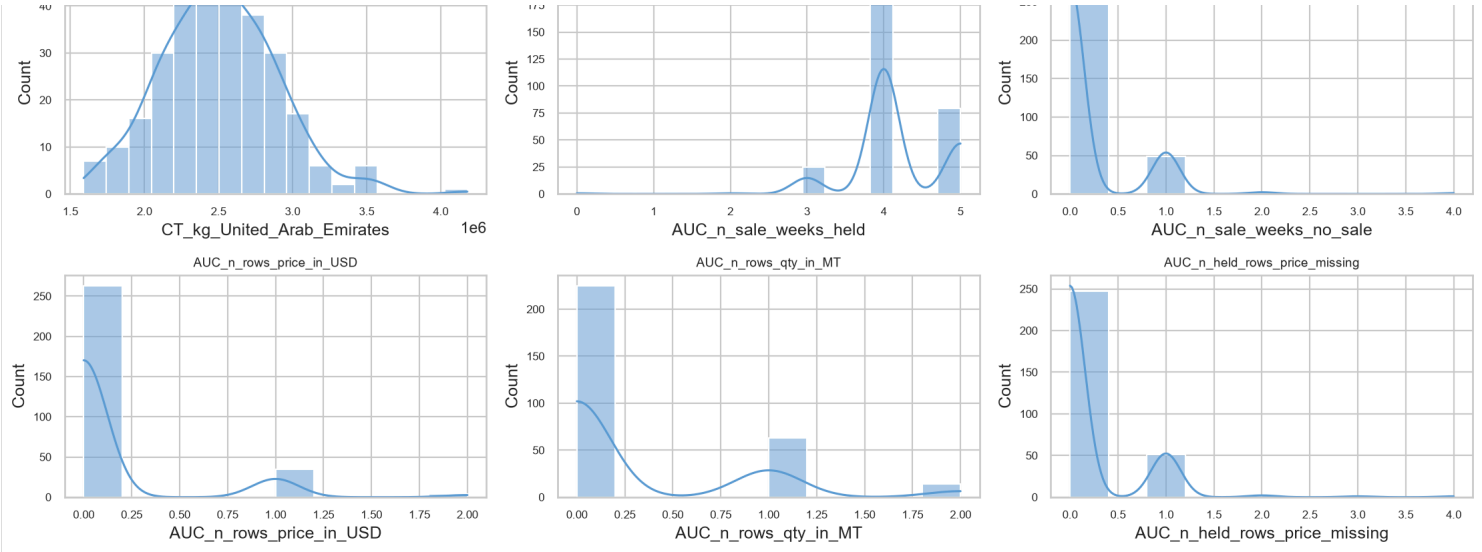
Missing Values



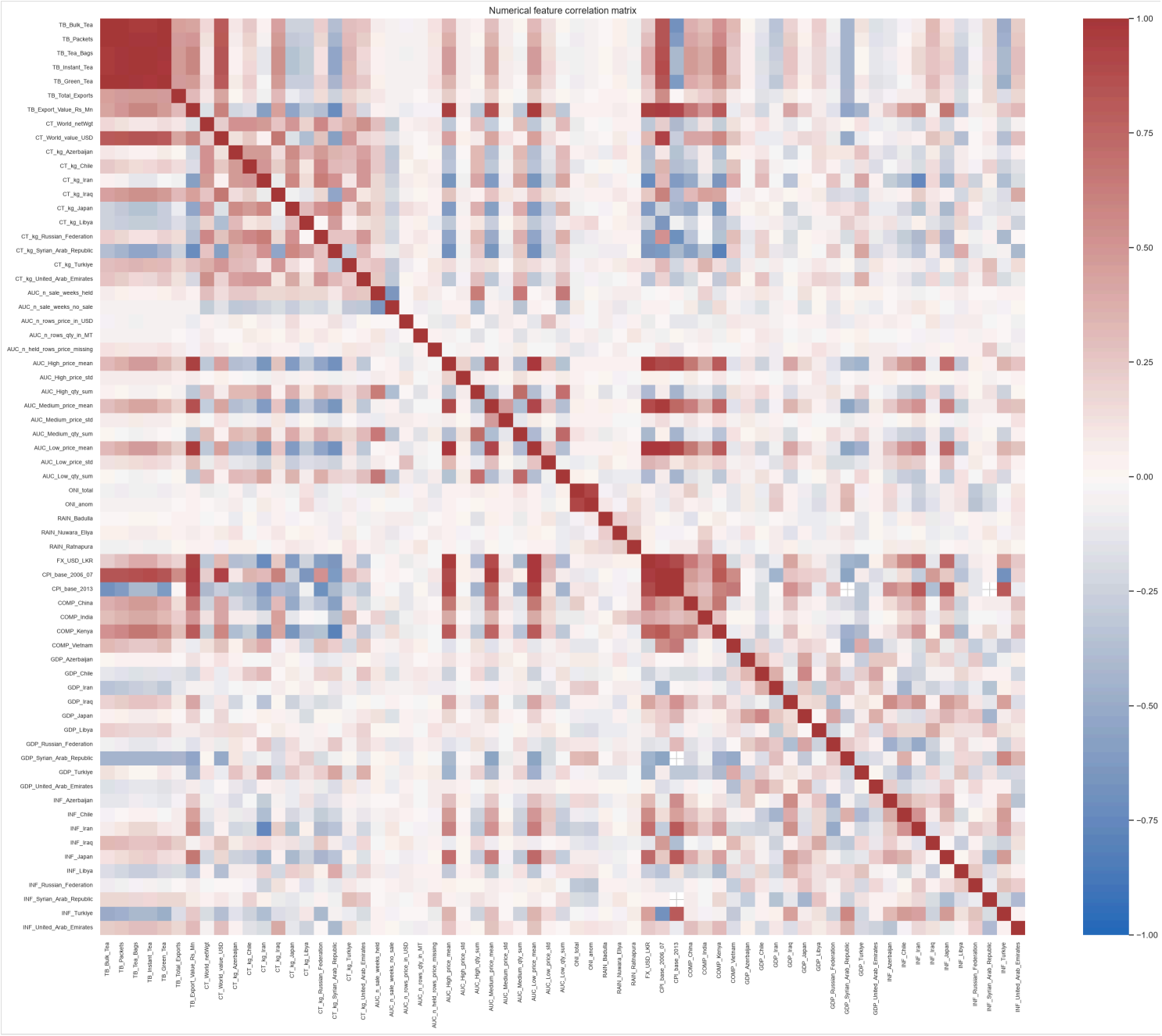
Numerical Distributions

Numerical feature distributions

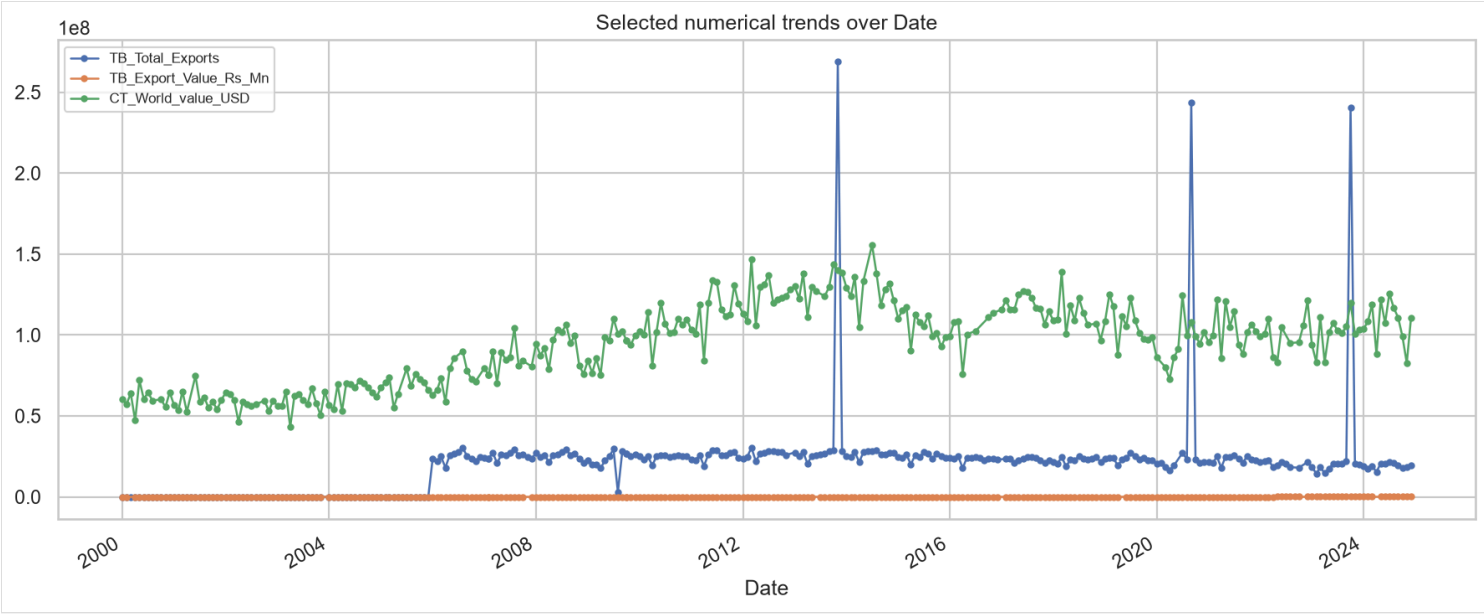




Correlation Heatmap



Relevant Dataset-Specific Chart


Selected numerical trends over Date
1e8
TB_Total_Exports
TB_Export_Value_Rs_Mn
CT_World_value_USD
2.5
2.0
1.5
1.0
0.5
0.0
2000
2004
2008
2012
2016
2020
2024
Date